

FACULTY OF COMPUTING

Sri Lanka Institute of Information Technology

BSc (Hons) in Data Science

IT3101 – Data Warehousing and Business Intelligence

Design and Implementation of a Data Warehouse and Business Intelligence Solution

Telecommunications Customer Churn Analysis

Y3.S1. WE. DS. 0201 - 2026-DS-10

Group Details:

Student Name	Student ID
Nirmani K.H.D.T.	IT24103380
Lakshitha Dilshan J.K.P	IT24102621
Palliyaguruge C.T	IT24103655
Senaratne P.A.R.T	IT24103526

Contents

Executive Summary 4

1. Task 1 – Dataset Selection and Business Scenario Identification 5

Detailed Attribute Dictionary 5

DWBI Suitability 7

2. Task 2 – Data Source Identification and Preparation 7

Data Quality and Edge Cases 8

3. Task 3 – Data Warehouse Architecture Design 9

4. Task 4 – Dimensional Data Warehouse Design and Implementation 10

Design Decision Matrix 14

Surrogate and Foreign Key Map 14

Future Slowly Changing Dimensions 14

5. Task 5 – ETL Process Development 15

Complete ETL Execution Sequence 15

6. Task 6 – Data Mart Development 16

Mart Object, Views, and Governance 17

7. Task 7 – OLAP Analysis and Business Intelligence Dashboard Development 17

Power BI Aggregations and OLAP Operations Matrix 19

8. Task 8 – Business Insights and Recommendations 20

Insight 1: Overall churn exposure 20

Insight 2: Risk-segment mix 20

Insight 3: Largest contract/service charge category 20

Insight 4: Join-quarter customer counts 20

Insight 5: Average charges by join cohort 20

Insight 6: City/contract churn-score contrast 21

Insight 7: Customer concentration by city 21

9. AI Usage Disclosure 21

10. References 21

Appendix A – Technical Evidence 21

Appendix B – Project File Inventory 22

Executive Summary

This project develops a telecom customer churn and retention analytics solution using 7,043 customer accounts. Python integrates customer/billing CSV, location-reference CSV, and service-subscription JSON sources, performs data preparation, derived-field creation and eight data-quality checks, and loads a SQLite warehouse containing a staging table, the star schema and a retention data mart. The dimensional design uses Fact_CustomerAccount with Customer, Location, Service, Contract, and Date dimensions, with a 73-row join-date Date dimension. The same tables are copied to SQL Server for Power BI.

The Mart_CustomerRetention data mart, built from the star schema by the ETL pipeline and exposed through six analytical views, supports retention analysis. The three-page Power BI dashboard presents executive KPIs, risk segments, contract/service charges, join-quarter comparisons, city/contract churn scores, and interactive customer analysis.

Dashboard-supported results include 26.54% overall churn, 2,559 High Risk customers, and $185,181.10 in summed monthly charges for the Month-to-month/Fiber optic treemap category.

1. Task 1 – Dataset Selection and Business Scenario Identification

The selected dataset is a telecom customer churn dataset representing an OLTP-style customer-account environment. Each row stores customer account, contract, billing, service, and churn outcome details, which is suitable for a dimensional data warehouse and analytical dashboard.

Business domain: Telecommunications / customer retention

Business problem addressed: identifying churn drivers, measuring revenue exposure, segmenting risky customers, and designing retention strategies.

Dataset source: IBM Telco Customer Churn dataset (extended version), sourced from Kaggle.

Dataset format: CSV / Excel-based operational data transformed into a warehouse-ready analytical dataset.

Number of records: 7,043 customer records.

The dataset is suitable for a DW/BI solution because it contains both factual measures such as Monthly Charges, Total Charges, Tenure Months, Churn Score, and CLTV, and descriptive attributes such as Gender, Contract, Payment Method, Technology Service, and City, which map naturally to dimension tables.

Source	Format	Records	Main content
Customer and billing system	CSV	7,043	Customer profile, billing, churn status, tenure, contract
Geographic reference system	CSV	1,652 unique ZIP codes	City, State, Country, latitude, longitude
Service provisioning system	JSON	7,043	Phone service, internet service, device protection, streaming, tech support

Detailed Attribute Dictionary

Attribute(s)	Data Type	DWBI Role	Description
CustomerID	Text / NVARCHAR(255)	Natural key	Source customer identifier; business key in Dim_Customer.
CustomerKey	Integer / INT IDENTITY	Surrogate key	Dim_Customer primary key; fact foreign key.
CustomerAccountKey	Integer / INT IDENTITY	Fact key	Identifies one account-grain row in the Task 4 fact export.
Gender, Senior_Citizen, Partner, Dependents	Text	Attributes	Customer demographic and household categories.

Tenure_Months	Integer	Measure	Account tenure in months.
Tenure_Years	Decimal	Derived measure	Tenure_Months divided by 12, rounded to one decimal place.
Contract	Text / NVARCHAR(100)	Attribute	Contract term category.
Paperless_Billing	Text	Attribute	Paperless billing status.
Payment_Method	Text / NVARCHAR(100)	Attribute	Payment method category.
Monthly_Charges	Decimal / DECIMAL(10,2)	Measure	Current monthly account charges.
Total_Charges	Decimal / DECIMAL(10,2)	Measure	Accumulated customer charges.
Avg_Monthly_Spend	Decimal / DECIMAL(10,2)	Derived measure	Total_Charges divided by tenure months; zero tenure is treated as one.
Churn_Label	Text	Outcome label	Text churn outcome in the customer/billing source.
Churn_Value	Integer / INT	Outcome / measure	Binary churn indicator; its average is used as a churn rate.
Churn_Score	Integer / INT	Risk score	Dataset-provided churn score; not a churn-rate percentage.
CLTV	Integer / INT	Measure / estimate	Customer lifetime value estimate.
Churn_Reason	Text	Outcome explanation	Recorded churn reason; ETL fills nulls with a not-applicable label.
Zip_Code	Integer in source; NVARCHAR(50) in SQL Server	Location attribute	ZIP identifier used to join the location reference.
City, State, Country	Text	Location attributes	Geographic categories from the location reference source.
Latitude, Longitude	Decimal / FLOAT	Location attributes	Geographic coordinates.
Phone_Service, Multiple_Lines	Text / NVARCHAR(20)	Service attributes	Phone and multiple-line service categories.
Internet_Service	Text / NVARCHAR(50)	Service attribute	Internet category such as DSL, Fiber optic, or No.
Online_Security, Online_Backup, Device_Protection, Tech_Support,	Text / NVARCHAR(20)	Service attributes	Service subscription categories.

Streaming_TV, Streaming_Movies			
LocationKey, ServiceKey, ContractKey, DateKey	Integer / INT IDENTITY	Surrogate keys	Dimension primary keys referenced by Fact_CustomerAccount.
FullDate	Date / DATE in SQL Server	Date attribute	Join date derived from tenure (snapshot date 2023-09-30 minus tenure × 30 days); used for the Year > Quarter > Month > Date hierarchy.
Year, MonthNumber	Integer	Date attributes	Numeric calendar hierarchy fields.
Quarter, MonthName, QuarterName	Text	Date attributes	Descriptive calendar hierarchy fields.
Risk_Segment	Text / NVARCHAR(50)	Risk attribute	High Risk, Medium Risk, or Low Risk value derived in the ETL from Churn_Score (Low <40, Medium 40–69, High ≥70) and carried into the fact table and the retention mart.
Join_Date, Join_Year, Join_Quarter	Date text, integer, text	Cohort attributes	Retention-mart cohort fields used to derive the separate 73-row Dim_Date export.

DWBI Suitability

- The 7,043 account-grain records provide a clear unit of analysis for churn and charge metrics.
- Numerical fields such as Monthly_Charges, Total_Charges, Tenure_Months, Churn_Value, Churn_Score, and CLTV support aggregation and comparison.
- Customer, contract, service, payment, and geographic attributes support dimensional slicing and grouping.
- The three source formats (CSV, CSV reference data, and JSON) demonstrate integration from multiple operational-style sources.
- The records support a fact-and-dimension model and a retention-focused mart for operational decision support.

2. Task 2 – Data Source Identification and Preparation

The implementation uses three operational-style source systems that represent a realistic architecture for reporting and analytical processing.

Source 1: Customer & Billing System (customer_billing_source.csv)

- This source stores demographic and account data such as CustomerID, Gender, Senior_Citizen, Partner, Dependents, Tenure_Months, Contract, Paperless_Billing, Payment_Method, Monthly_Charges, Total_Charges, Churn_Value, Churn_Score, CLTV, and Churn_Reason.

Source 2: Location Reference System (location_reference_source.csv)

- This lookup table stores ZIP-code-level geographic details including City, State, Country, Latitude, and Longitude. It enables dimensional analysis by location and supports geographical churn comparisons.

Source 3: Service Provisioning System (service_subscription_source.json)

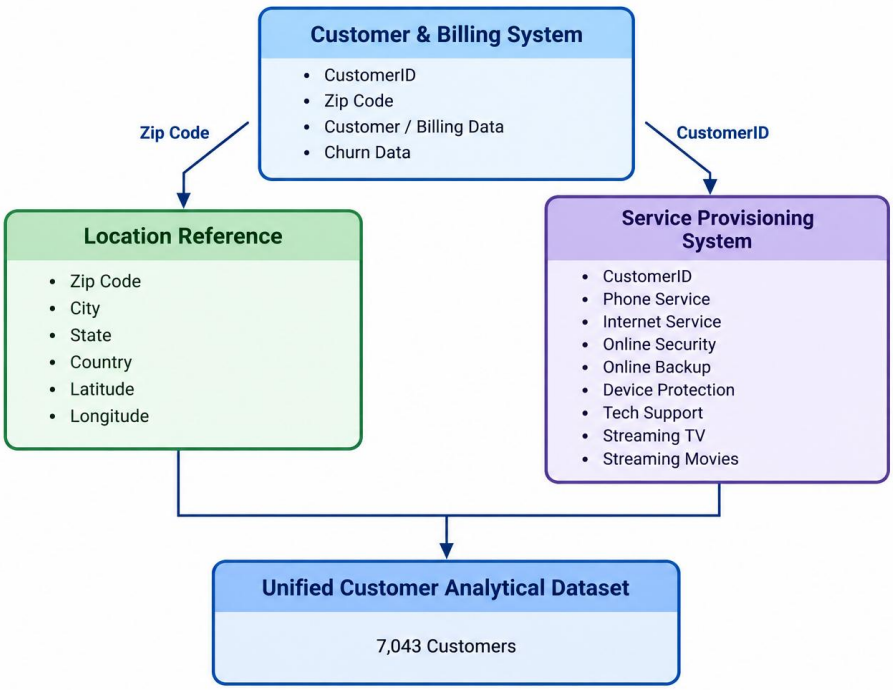
- This source contains service subscription features including Phone_Service, Multiple_Lines, Internet_Service, Online_Security, Online_Backup, Device_Protection, Tech_Support, Streaming_TV, and Streaming_Movies. It maps to the service dimension in the data warehouse.

Data preparation steps included loading each source into a staged analytical environment, trimming text values, filling blank Total_Charges values with 0, standardizing fields, removing duplicates where needed, and creating derived fields such as Avg_Monthly_Spend and Tenure_Years. This ensured that the files could be merged into a unified analytical model without losing business meaning

Data Quality and Edge Cases

The ETL code replaces blank Total_Charges strings with 0 before numeric conversion. Its recorded run found 0 remaining blanks, indicating the input CSV had no blank values at execution time. Churn_Reason nulls are filled with 'Not Applicable – Active Customer'; the run found 0 remaining nulls. Duplicate CustomerID values are checked; 0 duplicates were found, so the pipeline removed no rows in that run.

Text/object values are stripped of leading and trailing whitespace (22 columns in the logged run). Churn_Value, Churn_Score, and CLTV are cast to integers. Avg_Monthly_Spend, Tenure_Years, Join_Date (snapshot date 2023-09-30 minus tenure × 30 days) and Risk_Segment (Low <40, Medium 40–69, High ≥70) are derived. The cleaned merged dataset is then loaded into the staging table Stg_CustomerMaster (7,043 rows × 35 columns) before the dimensions and fact table are built.



3. Task 3 – Data Warehouse Architecture Design

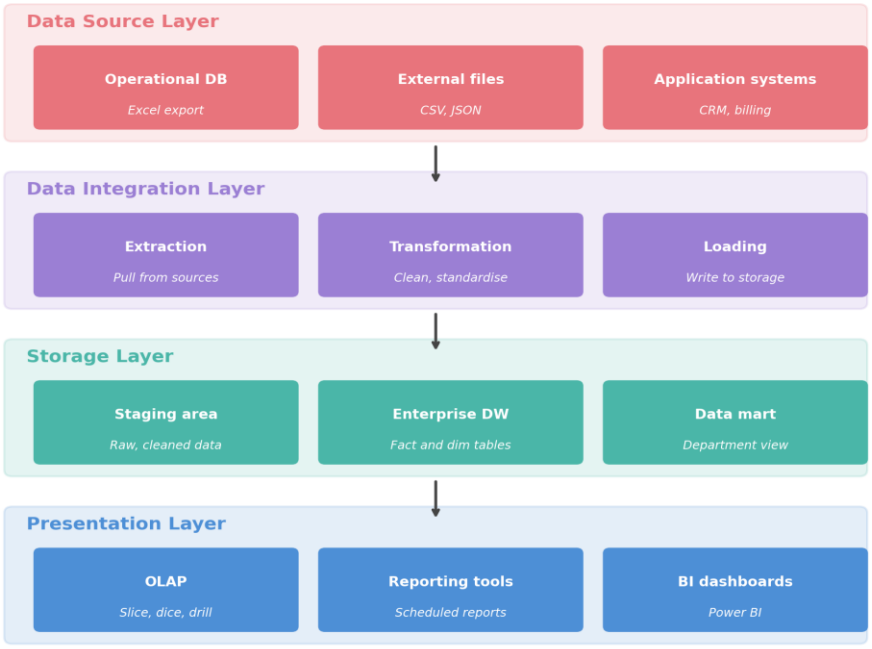
The solution follows an operational-to-analytical architecture with four major layers: Data Source Layer, Data Integration Layer, Storage Layer, and Presentation Layer.

Data Source Layer: operational sources including billing, location, and service systems.

Data Integration Layer: ETL pipeline built in Python that extracted, transformed, cleaned, standardised, and loaded the source data.

Storage Layer: staging data, normalized warehouse tables, and a retention-focused data mart for business analysis.

Presentation Layer: Power BI dashboard with executive summary, trend analysis, and interactive drill-down views.



Data flows top to bottom: sources feed integration, integration loads storage, storage powers dashboards

This architecture is suitable because it separates transactional systems from analytical reporting, allows data quality handling before load, and supports repeatable BI reporting with a single analytical source of truth.

4. Task 4 – Dimensional Data Warehouse Design and Implementation

The implementation uses a star schema with one fact table and multiple dimensions.

Fact table: Fact_CustomerAccount

Dimension tables: Dim_Customer, Dim_Location, Dim_Service, Dim_Contract, and Dim_Date

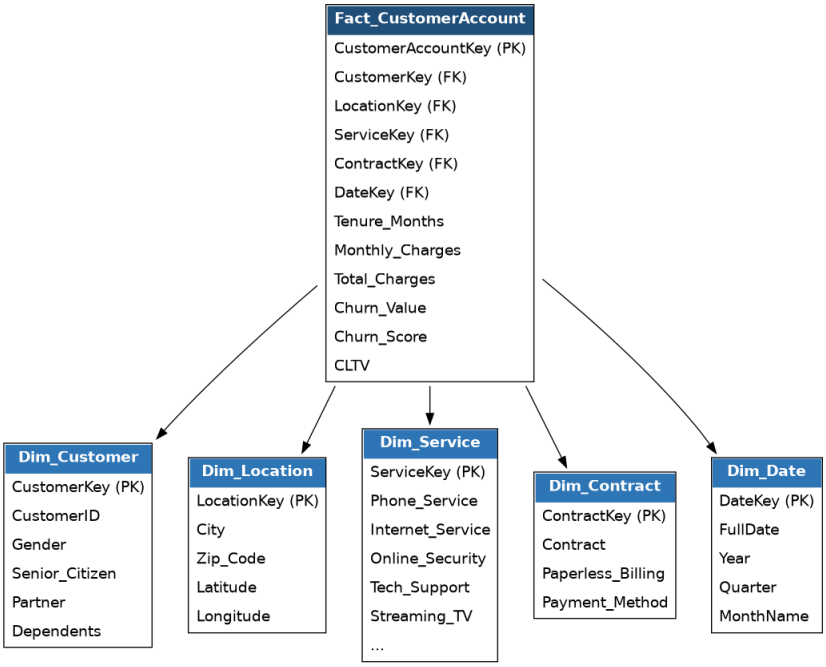
Business grain: one row per customer account record. The generated fact export contains 7,043 rows and one row per customer account, supporting churn and revenue analysis at customer-account level.

Fact key roles: CustomerAccountKey is the fact table primary key. CustomerKey, LocationKey, ServiceKey, ContractKey, and DateKey are foreign keys to their matching dimensions.

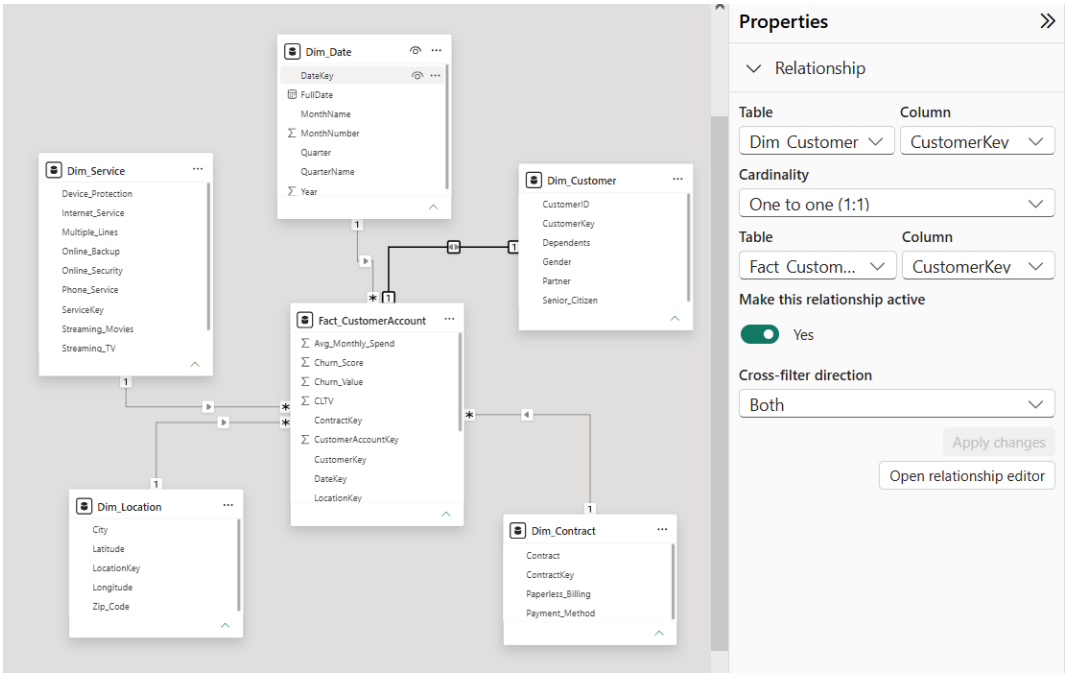
Measures and analysis attributes: Tenure_Months, Monthly_Charges, Total_Charges, Avg_Monthly_Spend, Tenure_Years, Churn_Value, Churn_Score, CLTV, and Risk_Segment. Churn_Value is a 0/1 indicator; its average represents the churn rate.

Dimension primary keys are CustomerKey, LocationKey, ServiceKey, ContractKey, and DateKey. The relationships are from each dimension key to its matching fact foreign key. The date hierarchy is Year > Quarter > Month, with MonthName sorted by MonthNumber. The customer key is unique in both the current fact and customer dimension because this dataset contains one account row per customer.

Date implementation note: the SQLite ETL and the SQL Server/Power BI export both use the same 73-row Dim_Date, derived from customer Join_Date values (2017-10-31 to 2023-09-30) and supporting quarterly analysis. Join_Date is derived from tenure and the 2023-09-30 snapshot date because the source system does not store it.



Power BI Model view evidence: the six fact and dimension tables are connected in the semantic model. Dim_Customer to Fact_CustomerAccount is an active one-to-one relationship because the dataset has one account row per customer.



SQL Server load validation:

SSMS row counts match the exported tables:

- Dim_Customer 7,043
- Dim_Location 1,652
- Dim_Service 322
- Dim_Contract 24
- Dim_Date 73
- Fact_CustomerAccount 7,043

A foreign-key validation query returned zero orphan fact rows for CustomerKey, LocationKey, ServiceKey, ContractKey, and DateKey (query and result shown below).

```
142  SELECT
143      (SELECT COUNT_BIG(*) FROM dbo.Dim_Customer) AS Dim_Customer,
144      (SELECT COUNT_BIG(*) FROM dbo.Dim_Location) AS Dim_Location,
145      (SELECT COUNT_BIG(*) FROM dbo.Dim_Service) AS Dim_Service,
146      (SELECT COUNT_BIG(*) FROM dbo.Dim_Contract) AS Dim_Contract,
147      (SELECT COUNT_BIG(*) FROM dbo.Dim_Date) AS Dim_Date,
148      (SELECT COUNT_BIG(*) FROM dbo.Fact_CustomerAccount) AS Fact_CustomerAccount;
149
150
```

2 % No issues found Ln: 142, Ch: 1 (408 chars, 7 lines) SPC

Results Messages

Dim_Customer	Dim_Location	Dim_Service	Dim_Contract	Dim_Date	Fact_CustomerAccount
7043	1652	322	24	73	7043

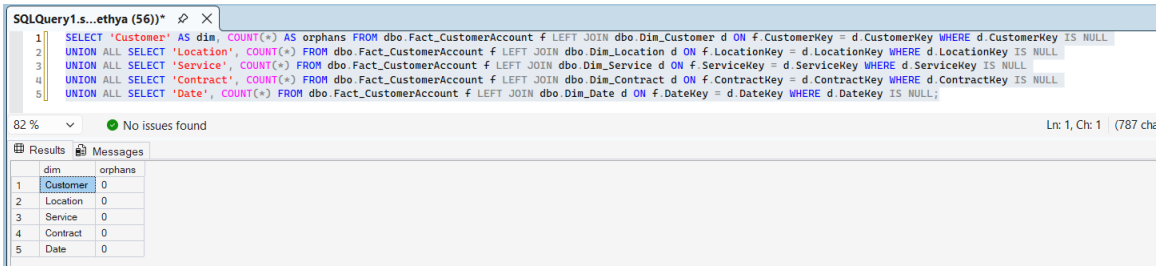


Table	Purpose	Key attributes
Fact_CustomerAccount	One row per customer account; PK CustomerAccountKey	FKs CustomerKey, LocationKey, ServiceKey, ContractKey, DateKey; Tenure_Months, Monthly_Charges, Total_Charges, Avg_Monthly_Spend, Tenure_Years, Churn_Value, Churn_Score, CLTV, Risk_Segment
Dim_Customer	Customer dimension; PK CustomerKey	CustomerID, Gender, Senior_Citizen, Partner, Dependents
Dim_Location	Location dimension; PK LocationKey	City, Zip_Code, Latitude, Longitude
Dim_Service	Service feature dimension; PK ServiceKey	Phone_Service, Multiple_Lines, Internet_Service, Online_Security, Online_Backup, Device_Protection, Tech_Support, Streaming_TV, Streaming_Movies
Dim_Contract	Contract and billing dimension; PK ContractKey	Contract, Paperless_Billing, Payment_Method
Dim_Date	Date reference dimension; PK DateKey	FullDate, Year, Quarter, MonthName, MonthNumber, QuarterName

The schema supports customer, service, location, and contract-based intelligence and provides a strong basis for executive summary, trend analysis, and segmentation reporting.

Design Decision Matrix

Decision	Implementation
Business process	Customer-account churn, retention, and charges analysis
Fact table	Fact_CustomerAccount
Grain	One row per customer account; 7,043 rows in the Task 4 export
Measures	Tenure_Months, Monthly_Charges, Total_Charges, Avg_Monthly_Spend, Tenure_Years, Churn_Value, Churn_Score, CLTV
Dimensions	Dim_Customer, Dim_Location, Dim_Service, Dim_Contract, Dim_Date

Surrogate and Foreign Key Map

Dimension Primary Key	Fact Foreign Key	Relationship Meaning
Dim_Customer.CustomerKey	Fact_CustomerAccount.CustomerKey	Customer account profile
Dim_Location.LocationKey	Fact_CustomerAccount.LocationKey	City and ZIP geography
Dim_Service.ServiceKey	Fact_CustomerAccount.ServiceKey	Subscribed service combination
Dim_Contract.ContractKey	Fact_CustomerAccount.ContractKey	Contract and payment attributes
Dim_Date.DateKey	Fact_CustomerAccount.DateKey	Join date derived from tenure; identical in the SQLite warehouse and the SQL Server export

The dimension keys are warehouse surrogate keys generated for the exports and declared as IDENTITY primary keys in the SQL Server DDL. CustomerID remains the natural/business identifier in Dim_Customer. In the current dataset CustomerKey is unique on both the dimension and fact sides because the grain is one account per customer.

Future Slowly Changing Dimensions

SCD processing is not implemented. If customer, location, service, or contract attributes change over time, a Type 2 design could add effective-start and effective-end dates plus a current-row indicator, create a new surrogate-key version for each change, and retain historical fact-to-dimension links.

5. Task 5 – ETL Process Development

The ETL process was implemented in Python using the pandas and sqlite3 libraries. It follows a reproducible workflow: **Extract → Transform → Load.**

Extract: the data was read from CSV and JSON sources, including the customer billing data, location reference table, and service subscriptions.

Transform: blank values were handled, text columns were standardised, duplicates were checked, numeric data types were validated, and derived attributes such as Avg_Monthly_Spend and Tenure_Years were created.

Load: transformed records were inserted into fact and dimension tables in SQLite, using surrogate keys to maintain relationship integrity and warehouse consistency.

The ETL log file documents each step and confirms the pipeline was run successfully. This provides evidence of orderly transformation and warehouse population.

```
C:\Users\Chamethya\Desktop\Y3S1\DATA WAREHOUSING AND BUSINESS INTELLIGENCE\ASSIGNMENT>py etl_pipeline.py
[22:43:25] EXTRACT: Reading Source 1 - Customer & Billing System (CSV)
[22:43:25]   -> 7043 rows, 17 columns extracted
[22:43:25] EXTRACT: Reading Source 2 - Location Reference System (CSV)
[22:43:25]   -> 1652 rows, 6 columns extracted
[22:43:25] EXTRACT: Reading Source 3 - Service Provisioning System (JSON)
[22:43:25]   -> 7043 rows, 10 columns extracted
[22:43:25] TRANSFORM: Joining sources on Zip_Code and CustomerID
[22:43:25]   -> Merged dataset: 7043 rows, 31 columns
[22:43:25] TRANSFORM: Missing value handling
[22:43:25]   -> Fixed 0 blank Total_Charges values (set to 0, cast to float)
[22:43:25]   -> Filled 0 null Churn_Reason values
[22:43:25] TRANSFORM: Deduplication check
[22:43:25]   -> Duplicate CustomerIDs found: 0 (none removed - already clean)
[22:43:25] TRANSFORM: Standardisation - trimming whitespace on text columns
C:\Users\Chamethya\Desktop\Y3S1\DATA WAREHOUSING AND BUSINESS INTELLIGENCE\ASSIGNMENT\etl_pipeline.py:70: Pandas4Warning: For backward compatibility, 'str' dtypes are included by select_dt
ypes when 'object' dtype is specified. This behavior is deprecated and will be removed in a future version. Explicitly pass 'str' to `include` to select them, or to `exclude` to remove the
m and silence this warning.
See https://pandas.pydata.org/docs/user_guide/migration-3-strings.html#string-migration-select-dtypes for details on how to write code that works with pandas 2 and 3.
  text_cols = merged.select_dtypes(include='object').columns
[22:43:25]   -> Trimmed 22 text columns
[22:43:25] TRANSFORM: Derived attribute creation
[22:43:25]   -> Created Avg_Monthly_Spend and Tenure_Years derived columns
[22:43:25] TRANSFORM: Data type conversion check
[22:43:25]   -> Confirmed Churn_Value, Churn_Score, CLTV as integer types
[22:43:25] TRANSFORM COMPLETE: Final cleaned dataset 7043 rows x 33 columns
[22:43:25] LOAD: Connecting to target warehouse (SQLite)
[22:43:25] LOAD: Building Dim_Customer
[22:43:25]   -> Loaded 7043 rows
[22:43:25] LOAD: Building Dim_Location
[22:43:25]   -> Loaded 1652 rows
[22:43:25] LOAD: Building Dim_Service (junk dimension)
[22:43:25]   -> Loaded 322 rows
[22:43:25] LOAD: Building Dim_Contract
[22:43:25]   -> Loaded 24 rows
[22:43:25] LOAD: Building Dim_Date
[22:43:25]   -> Loaded 1 row
[22:43:25] LOAD: Building Fact_CustomerAccount (attaching surrogate keys)
[22:43:25]   -> Loaded 7043 rows into Fact_CustomerAccount
[22:43:25] VALIDATION: Checking for null foreign keys
[22:43:25]   -> Null foreign keys found: 0 (expected: 0)
[22:43:25] VALIDATION: Comparing churn rate before vs after load
[22:43:25]   -> Source churn rate: 26.5370% | Warehouse churn rate: 26.5370% | Match: True
[22:43:25] VALIDATION: Row count reconciliation
[22:43:25]   -> Source rows: 7043 | Fact table rows: 7043 | Match: True
[22:43:25] ETL PIPELINE COMPLETE
```

ETL validation checks included eight data-quality rules (seven passed with zero violations; one non-critical tolerance check flagged two records, which were retained), a null foreign-key check, an orphan check of every fact foreign key against its dimension (zero orphan rows), churn-rate matching before and after the load (26.5370%), row-count reconciliation between the source, staging table, fact table and mart (all 7,043), and reconciliation of the mart views to the mart.

Complete ETL Execution Sequence

1. Extract customer/billing CSV (7,043 rows), location-reference CSV (1,652 rows), and service-subscription JSON (7,043 rows). The raw Excel export (7,043 rows × 33 columns) is also read to audit source preparation: CustomerID sets are identical across sources and all 11 blank Total Charges belong to tenure-0 accounts.

2. Left-join location on Zip_Code and service subscriptions on CustomerID; clean blanks, trim text, check duplicate CustomerID values, convert numeric types, derive Avg_Monthly_Spend, Tenure_Years, Join_Date and Risk_Segment, and apply the eight data-quality rules.
3. Load the cleaned merged dataset into the staging table Stg_CustomerMaster (7,043 rows × 35 columns).
4. Build dimensions and assign sequential surrogate keys to distinct dimension rows.
5. Load SQLite dimensions in this order: Dim_Customer, Dim_Location, Dim_Service, Dim_Contract, Dim_Date; then attach keys and load Fact_CustomerAccount; then build Mart_CustomerRetention from the fact and dimension tables and create the six retention views.
6. Copy the five dimensions and the fact table from telecom_dw.db to SQL Server with load_to_sqlserver.py (dimensions first, Fact_CustomerAccount last), so Power BI and SSMS use exactly the ETL output, including the 73-row Dim_Date.
7. Validate null foreign keys, compare churn rate before/after load, reconcile source/fact counts, and check SQL Server for unmatched fact keys.

The SQLite ETL (etl_pipeline.py) writes the staging table, all five dimensions, the fact table, Mart_CustomerRetention and its views to one database, telecom_dw.db, and exports retention_mart.csv. The SQL Server load copies these same tables, so the SQLite and SQL Server outputs are one synchronized load.

6. Task 6 – Data Mart Development

A department-specific data mart was created to support retention and churn analysis for business users. The mart is centred on customer risk and churn exposure and is designed for decision-makers in customer retention and commercial strategy.

Purpose: support proactive risk management and customer retention planning.

Target users: customer success managers, sales and retention managers, executive leadership, and telecom operations teams.

Analytical benefits: the data mart enables churn segmentation, contract-risk analysis, service sensitivity analysis, and monthly revenue exposure reviews.

The final retention mart includes fields such as CustomerID, Contract, Internet_Service, Payment_Method, Monthly_Charges, Churn_Value, Risk_Segment, and Join_Quarter, which make it directly useful for BI dashboards and retention planning.

Mart Object, Views, and Governance

The implemented mart object is the physical table Mart_CustomerRetention in telecom_dw.db, with 7,043 rows and 29 columns, built by the ETL pipeline from the fact and dimension tables; retention_mart.csv is its flat-file export.

Six analytical views sit on the mart: vw_Retention_Summary (headline KPIs, 1 row), vw_Retention_ByRiskSegment (3 rows), vw_Retention_ByContract (contract × internet service, 9 rows), vw_Retention_ByTenureBand (4 rows), vw_Retention_ByJoinQuarter (24 rows) and vw_AtRisk_ActiveCustomers (950 active High Risk customers, the retention team's call list). The view totals reconcile to the 7,043 customers in the mart, and retention_mart_sqlserver.sql recreates the mart and views in SQL Server. The mart's curated, denormalized fields reduce query complexity for retention analysis.

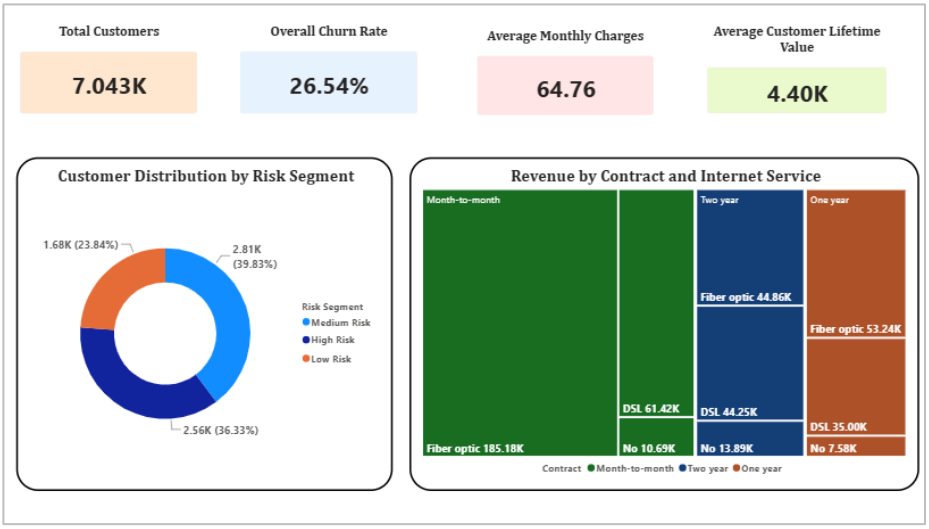
A dedicated mart simplifies user access and supports consistent definitions. Access control such as SQL roles or row-level security is not configured in this project and would be applied at deployment.

7. Task 7 – OLAP Analysis and Business Intelligence Dashboard Development

The solution uses Microsoft Power BI to present an interactive warehouse-based dashboard with three report pages. The final star-schema report is saved as **Telecom_Churn_Dashboard_StarSchema.pbix**; its visual field bindings reference Fact_CustomerAccount and the applicable dimensions rather than retention_mart.

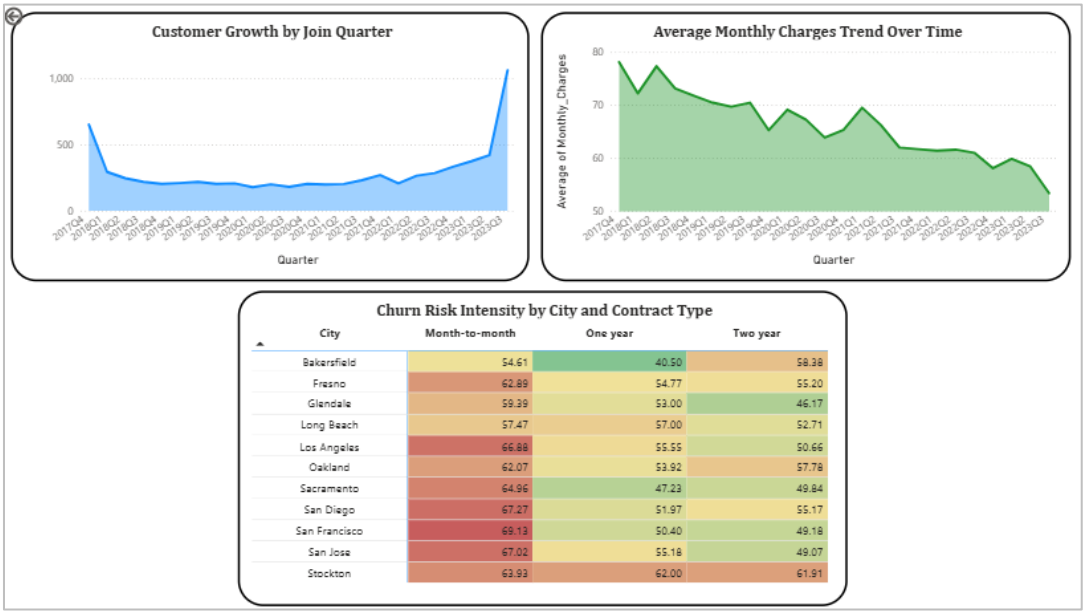
Report Page 1: Executive Summary

This page contains KPI cards for 7,043 customers, 26.54% overall churn, $64.76 average monthly charges, and $4,400 average customer lifetime value. A donut chart shows the customer distribution by Risk_Segment, and a treemap shows summed monthly charges by Contract and Internet_Service.



Report Page 2: Trend Analysis

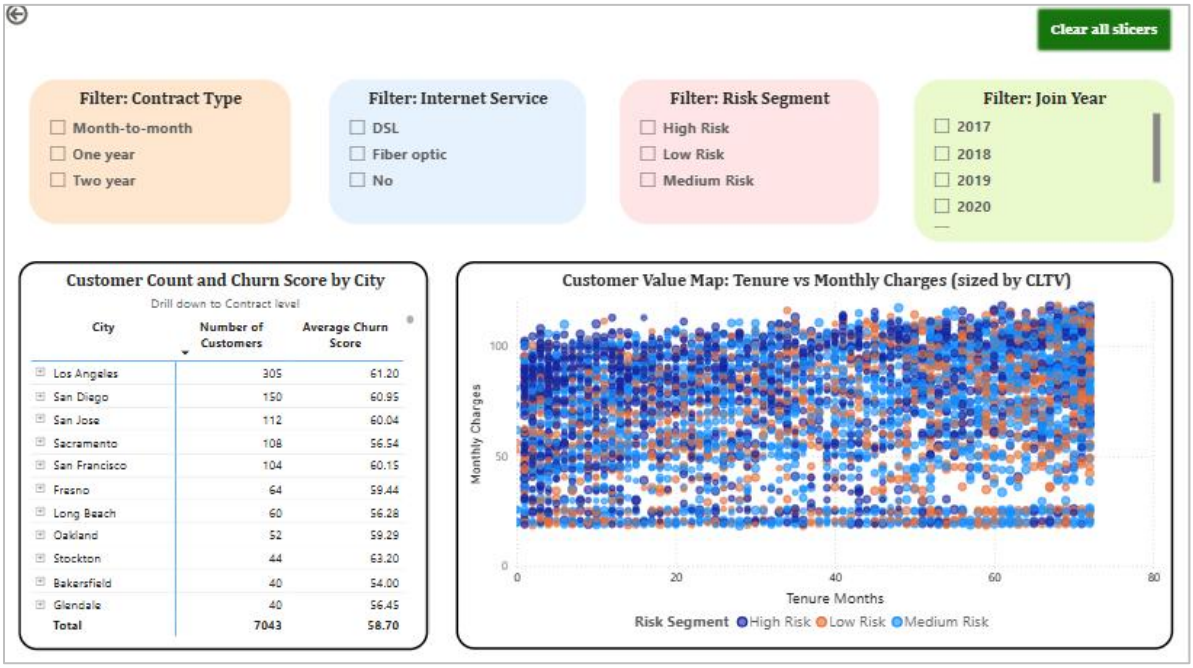
This page compares customer counts by Join_Quarter and average Monthly_Charges by Join_Quarter. A city-by-contract matrix compares average Churn_Score for the displayed cities; the Top N city filter ranks cities by customer count. Churn_Score is a score, not a churn-rate percentage.



City	Month-to-month	One year	Two year
Bakersfield	54.61	40.50	58.38
Fresno	62.89	54.77	55.20
Glendale	59.39	53.00	46.17
Long Beach	57.47	57.00	52.71
Los Angeles	66.88	55.55	50.66
Oakland	62.07	53.92	57.78
Sacramento	64.96	47.23	49.84
San Diego	67.27	51.97	55.17
San Francisco	69.13	50.40	49.18
San Jose	67.02	55.18	49.07
Stockton	63.93	62.00	61.91

Report Page 3: Interactive Analysis

This page provides Contract, Internet_Service, Risk_Segment, and Year slicers, a clear-all-slicers control, a city/contract matrix with customer counts and average Churn_Score, and a customer-level scatter plot of Tenure_Months against Monthly_Charges, sized by CLTV and colored by Risk_Segment. The city/contract matrix supports drill-down and roll-up between city and contract levels.



City	Number of Customers	Average Churn Score
Los Angeles	305	61.20
San Diego	150	60.95
San Jose	112	60.04
Sacramento	108	56.54
San Francisco	104	60.15
Fresno	64	59.44
Long Beach	60	56.28
Oakland	52	59.29
Stockton	44	63.20
Bakersfield	40	54.00
Glendale	40	56.45
Total	7043	58.70

Power BI Aggregations and OLAP Operations Matrix

The visuals use implicit aggregations; the relationship guide's sample formulas are recommendations.

Visual Metric	PBIX Field(s)	PBIX Aggregation
Total customers	Fact_CustomerAccount[CustomerAccountKey]	Count of nonblank values; implicit visual aggregation
Overall churn rate	Fact_CustomerAccount[Churn_Value]	Average of the 0/1 outcome
Average monthly charges	Fact_CustomerAccount[Monthly_Charges]	Average
Average CLTV	Fact_CustomerAccount[CLTV]	Average
Customers by risk segment	Fact_CustomerAccount[Risk_Segment] and [CustomerAccountKey]	Category by count of nonblank account keys
Treemap charges	Fact_CustomerAccount[Monthly_Charges]	Sum by Dim_Contract[Contract] and Dim_Service[Internet_Service]; Average Churn_Value tooltip
Join-quarter customers	Fact_CustomerAccount[CustomerAccountKey] by Dim_Date[Quarter]	Count of nonblank account keys
Join-quarter charges	Fact_CustomerAccount[Monthly_Charges] by Dim_Date[Quarter]	Average
City/contract score	Fact_CustomerAccount[Churn_Score] by City and Contract	Average; City Top 10 is ranked by account count
Customer scatter	Tenure_Months, Monthly_Charges, CLTV, Risk_Segment, CustomerAccountKey	Average X/Y/Size, Risk_Segment series, account key identifies customer-level points

OLAP Operation	Existing Dashboard Control
Slice	Select one Contract, Internet_Service, Risk_Segment, or Year slicer value.
Dice	Combine two or more slicers to filter a multi-attribute subset.
Drill-down	Expand the Interactive Analysis matrix from City to Contract rows.
Roll-up	Collapse the matrix from Contract level back to City.
Cross-filter	Slicer and visual selections filter or highlight related report visuals.

The dashboard demonstrates OLAP-style analysis by enabling business users to compare dimensions and fact measures efficiently, which is a key objective of the assignment.

8. Task 8 – Business Insights and Recommendations

Insight 1: Overall churn exposure

- Technical Evidence: 1,869 of 7,043 accounts have Churn_Value = 1, giving an overall churn rate of 26.54%.
- Actionable Recommendation: Use the Executive Summary churn KPI as the baseline for monitoring retention initiatives.

Insight 2: Risk-segment mix

- Technical Evidence: The risk donut shows 2,559 High Risk customers (36.33%), 2,805 Medium Risk customers (39.83%), and 1,679 Low Risk customers (23.84%).
- Actionable Recommendation: Prioritize review of High Risk accounts and monitor the larger Medium Risk segment.

Insight 3: Largest contract/service charge category

- Technical Evidence: The treemap's Month-to-month/Fiber optic category has $185,181.10 in summed monthly charges, the largest displayed category.
- Actionable Recommendation: Review retention offers and service experience for this high-charge cohort; the visual shows revenue exposure, not proof of churn causality.

Insight 4: Join-quarter customer counts

- Technical Evidence: The cohort chart shows 419 accounts in 2023Q2 and 1,062 in 2023Q3. Join_Date is derived from tenure/snapshot data, so this is not a verified acquisition time series.
- Actionable Recommendation: Validate the apparent Q3 cohort increase against acquisition records before using it for staffing or growth decisions.

Insight 5: Average charges by join cohort

- Technical Evidence: Average Monthly_Charges are $78.09 for 2017Q4 and $53.32 for 2023Q3 join cohorts, a 31.72% difference. This compares cohorts, not customer prices over time.
- Actionable Recommendation: Compare plan mix and customer value across cohorts before changing pricing or offers.

Insight 6: City/contract churn-score contrast

- Technical Evidence: The displayed matrix reports average Churn_Score 69.13 for San Francisco/Month-to-month and 40.50 for Bakersfield/One year; the Bakersfield cell has six accounts. Churn_Score is not a churn-rate percentage.
- Actionable Recommendation: Review higher-scoring combinations and validate them against actual churn outcomes before targeting interventions.

Insight 7: Customer concentration by city

- Technical Evidence: The displayed city matrix shows 305 Los Angeles accounts and 150 San Diego accounts.
- Actionable Recommendation: Prioritize investigation of customer experience and retention patterns in the largest displayed city cohorts.

9. AI Usage Disclosure

AI tools were used as a supporting aid for brainstorming, report wording improvement, and refinement of analytical explanations. All final technical work, ETL logic, and business interpretations were reviewed and validated by the author against the project data and assignment requirements. The final submission reflects the author's own understanding and implementation decisions.

10. References

- IBM Telco Customer Churn Dataset (extended version), Kaggle.
- Power BI Dashboard implementation for customer retention and churn analysis.
- Python ETL pipeline for data extraction, cleaning, and warehouse loading.
- Assignment brief: IT3101 – Data Warehousing and Business Intelligence.

Appendix A – Technical Evidence

Evidence File(s)	Description
etl_script_screenshot.png	Python ETL implementation.
etl_log_screenshot.png; etl_log.txt	ETL execution stages and logged row-count/churn validation.
star_schema_erd.png	Designed star schema for Task 4.
insights and dashboard pages/semantic_model_star_schema.png	Power BI Model view with the fact and five dimension tables and active relationships.
insights and dashboard pages/ssms_row_counts.png	SSMS table counts: 7,043 customer, 1,652 location, 322 service, 24 contract, 73 date, and 7,043 fact rows.

insights and dashboard pages/orphan.png	SQL Server foreign-key orphan query returning zero orphan rows for all five dimensions.
insights and dashboard pages/page1_executive_summary.png; page2_trend_analysis.png; page3_interactive_analysis.png	The three Power BI report pages.
insights and dashboard pages/filtering_before.png; filtering_after.png	Filter interaction evidence.
insights and dashboard pages/drilldown_before.png; drilldown_after.png	Drill-down interaction evidence.
insights and dashboard pages/brushing_before.png; brushing_after.png	Visual-selection/brushing evidence.
insights and dashboard pages/architecture_diagram.png; Source Integration and Relationship Diagram.png	Architecture and source-integration diagrams.

The orphan-check query was run in SQL Server Management Studio; each of the five dimensions returned zero orphan fact rows (orphan.png).

Appendix B – Project File Inventory

File or Folder	Description
IT3101_DWBI_Report_Final_(2026-DS-10)	Final report
Telecom_Churn_Dashboard_StarSchema.pbix	Final three-page Power BI report using the fact and dimension tables.
telecom_dw.db	Populated SQLite warehouse: Stg_CustomerMaster, five dimensions, Fact_CustomerAccount, Mart_CustomerRetention and six retention views.
retention_mart.csv	7,043-row retention mart export.
customer_billing_source.csv	Customer and billing source data.
location_reference_source.csv	ZIP-code location reference source.
service_subscription_source.json	Service subscription source data.
Telco_customer_churn.xlsx	Original customer churn workbook.
etl_pipeline.py	Python/pandas/SQLite extraction, transformation, and load implementation.
etl_log.txt	ETL run and validation log.
etl_script_screenshot.png; etl_log_screenshot.png	ETL implementation and execution screenshots.

build_final_assignment_report.py	Reproducible report and summary generator.
generate_ssms_star_schema.py	Generates Task 4 star-schema SQL and CSV exports.
assignment_completion_summary.md	Submission contents and implementation notes.
star_schema_erd.png	Star-schema diagram.
insights and dashboard pages/architecture_diagram.png	High-level DW/BI architecture.
insights and dashboard pages/Source Integration and Relationship Diagram.png	Source integration diagram.
insights and dashboard pages/page1_executive_summary.png; page2_trend_analysis.png; page3_interactive_analysis.png	Dashboard page screenshots.
insights and dashboard pages/filtering_before.png; filtering_after.png; drilldown_before.png; drilldown_after.png; brushing_before.png; brushing_after.png	Interactive filtering, drill, and brushing evidence.
insights and dashboard pages/semantic_model_star_schema.png	Power BI semantic model screenshot.
insights and dashboard pages/ssms_row_counts.png	SSMS imported row-count evidence.
ssms_powerbi_star_schema/ssms_star_schema.sql	SQL Server star-schema DDL with primary and foreign keys.
ssms_powerbi_star_schema/Fact_CustomerAccount.csv	Task 4 account-grain fact export.
ssms_powerbi_star_schema/Dim_Customer.csv; Dim_Location.csv; Dim_Service.csv; Dim_Contract.csv; Dim_Date.csv	Task 4 dimension exports.
ssms_powerbi_star_schema/powerbi_relationship_guide.md	Power BI relationship and import guidance.